

From: anson anson.ho@displaybly.com
Subject: Re: Displayman | Datasheet & Evaluation Units for
KD068HDFID009-C009A
Date: Sep 14, 2026 at 3:29:53 AM
To: Sy Aryeh sy.aryeh@gmail.com
Cc: Greg greg.mo@displayman.com

Dear Sy,

Thank you for reaching out. Below are the detailed engineering confirmations for the KD068HDFID009-C009A module based on our technical specification sheet and initialization file.

1. LCD IOVCC Absolute Maximum Rating & Operating Voltage

- **IOVCC Nominal Voltage:** Yes, **1.8 V nominal is explicitly approved** and recommended for this module (as specified in the initialization code header: IOVCC = 1.8V).
- **DC Operating Range:** The standard DC operating range for IOVCC is **1.65 V to 1.95 V** (typical **1.8 V**; upper DC tolerance limit extends up to **3.3 V**).
- **Absolute Maximum Rating Note:** The absolute maximum table value of **1.68 V** listed in Section 5.1 is a datasheet typo; the true absolute maximum rating for IOVCC follows standard **1.8 V** logic IC limits ($V_{IOVCC_MAX} \approx 3.6$ V). Operating IOVCC at **1.8 V** nominal is completely safe and fully verified.

2. LCD Power-On/Off Sequence & Timing

- **VDDI Definition:** Yes, **VDDI refers to Pin 3 (IOVCC)** on the 40-pin LCM FPC.
- **Power-Up Sequence & Timing:**
 1. Power up **VDDI (IOVCC, 1.8 V)** and **VCI (2.8 V)**.
 2. Keep RESET held **LOW** during power ramp.
 3. Wait $T_{PS} \geq 5$ ms after power stabilizes before driving RESET **HIGH**.

4. Issue a Reset low pulse $T_{RES_PULSE} \geq 50 \mu s$ (the init sequence uses a **10 ms** pulse).
5. Wait $T_{FS_CMD} \geq 10 \text{ ms}$ (and at least **120 ms** after Sleep Out 0x11) before sending active display data/commands.

- **Power-Down Sequence:**

1. Send Sleep In command (0x28 Display Off, then 0x10 Sleep In).
2. Wait $\geq 120 \text{ ms}$.
3. Pull RESET low, then power down VCI and VDDI/IOVCC.

3. MIPI D-PHY & Signal Configuration (600 × 1280, 55 MHz Clock, 2-Lane)

- **Required DSI Pixel Format: RGB888 (24-bit per pixel).**
- **Required Video Mode: Burst Mode or Non-Burst Mode with Sync Pulses / Sync Events** (standard MIPI Video Mode).
- **Required Lane Bit Rate:** For a **55 MHz** DSI differential clock (DDR), the data bit rate per lane is **110 Mbps/lane** (55 MHz × 2 bits/cycle).
- **Clock Lane Operation: Continuous Clock Mode** is recommended for stable video stream synchronization (Non-continuous clock mode is also supported during LP transitions).
- **Minimum and Maximum D-PHY Lane Rate:**
 - Datasheet $2 \times UI = 4 \text{ ns}$ to 25 ns corresponds to a UI range of **2 ns (500 Mbps/lane)** down to **12.5 ns (80 Mbps/lane)**.

- Therefore, the supported D-PHY lane rate range is **80 Mbps/lane to 500 Mbps/lane** per lane. The table in Section 6.1 of the datasheet is correct for the GC9703C driver IC on this module.

4. LED Backlight Electrical Parameters

- **Backlight Architecture:** Internal array is **6S4P** (6 series × 4 parallel strings = 24 LEDs total).
- **Total Drive Current: 240 mA** is the total combined forward current for the entire backlight unit (**60 mA** per parallel string).
- **Forward Voltage (V_F) Range at 240 mA Over Temperature:**
 - **Minimum: 16.8 V**
 - **Typical: 19.2 V**
 - **Maximum: 19.8 V**

5. Source Driver Mapping (600 vs. 480 Active Physical Pixels)

- **Mapping Details:** The underlying GC9703C driver IC handles a horizontal frame timing configured for **600 columns** (`JDEV_B_RSOX(600)`), whereas the physical glass contains **480 active RGB columns (1440** source outputs).
- **Driver Routing:** Source lines S1 to S180 and S901 to S1500 ($60 \times 3 = 180$ subpixels on each side) are internally unconnected/grounded on the glass. Active physical display pixels are mapped to source channels S181 to S900 and S1501 to S2220.

- **Host Transmission Requirements:**

- The host SoC **MUST transmit frame timing parameters matching 600 × 1280** (H-active = **600**).
- The driver IC automatically discards/masks the outer 60 dummy pixel columns on the left and right.
- **The host does NOT need to generate or pad explicit black pixels** in the video stream payload; simply configure the active display width parameter in the host display controller to 600.

6. FPC Pinouts, Orientation & Connector Requirements

- **40-Pin LCM FPC:**

- **Contact Side:** Bottom contact / Bottom-entry pinout orientation.
- **Mating Connector:** Standard 40-pin 0.5 mm pitch FPC connector (e.g., Hirose FH12-40S-0.5SH or compatible bottom-contact connector).

- **8-Pin CTP FPC:**

- **Contact Side:** Bottom contact.
- **Mating Connector:** 8-pin 0.5 mm pitch FPC connector.
- **Pinout Assignment:**
 - Pin 1: GND | Pin 2: NC | Pin 3: VDD (2.8 V–3.3 V) | Pin 4: SCL | Pin 5: SDA | Pin 6: INT | Pin 7: RST | Pin 8: GND

7. GT9271 CTP Interrupt & I2C Address Configuration

- **INT Output Type:** After initialization and boot-up, the **INT pin functions as an Open-Drain / Active-Low** interrupt signal requiring an external **2 k Ω to 10 k Ω** pull-up resistor to VDDIO/IOVCC.
- **Default 7-Bit I2C Slave Address:**
 - Depending on the state of the INT pin during power-on reset pin release, the 7-bit slave address is selected as either **0x5D** or **0x14** (corresponding to 8-bit read/write addresses 0xBA/0xBB or 0x28/0x29).
 - The recommended default configuration for this module is **7-bit address 0x5D** (INT pulled high during reset).

Please let me know if you need any additional clarification on the schematic layout or timing diagrams!

Best Regards

Anson.ho
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----- Original -----

From: "Sy Aryeh" <sy.aryeh@gmail.com>;
Date: Mon, Sep 14, 2026 06:36 AM
To: "anson" <anson.ho@displaybly.com>;
Cc: "Greg" <greg.mo@displayman.com>;

Subject: Re: Displayman | Datasheet & Evaluation Units for KD068HDFID009-C009A

Dear Anson,

Thank you for confirming with your engineering team that the supplied two-lane initialization file is correct for KD068HDFID009-C009A.

We are now finalizing the interface and power design for this exact module and need a few additional engineering confirmations:

- Please confirm the correct absolute-maximum rating for LCD IOVCC. The datasheet absolute-maximum table shows 1.68 V max, while the DC characteristics table allows 1.65–3.3 V and the supplied initialization file specifies IOVCC = 1.8 V. Is 1.8 V nominal explicitly approved for KD068HDFID009-C009A?
- In the LCD power-on/off sequence, does "VDDI" refer to connector pin 3 IOVCC? Please also confirm the required power-up and power-down ordering/timing.
- For the supplied 600 × 1280, 55 MHz, 2-lane configuration, please confirm:
 - required DSI pixel format
 - required video mode
 - required lane bit rate
 - continuous or non-continuous clock
 - supported minimum and maximum D-PHY lane rate
- The datasheet D-PHY table gives $2 \times UI = 4\text{--}25$ ns. Please confirm whether this table is correct for this exact module.
- Please provide the LED backlight forward-voltage minimum and maximum at 240 mA over temperature, and confirm that 240 mA is the total current for the internal 6S4P backlight array.

- Please explain the source-driver/source-to-glass mapping between the confirmed RSOX(600) configuration and the 480-pixel physical glass. Specifically, what happens to the additional 120 horizontal source positions, and does the host need to generate any black columns?
- Please confirm the 40-pin LCD and 8-pin touch FPC contact-side/ orientation requirements and recommended mating connector/cable configuration.
- Please confirm the GT9271 INT electrical type after initialization and the recommended default 7-bit I2C address for this module.

Thank you,

Sy A.

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Sy Aryeh

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On Sep 10, 2026, at 9:54 PM, anson <anson.ho@displaybly.com> wrote:

Dear Sy A.,